Creating Slicers, Tables, and Trend Line Visuals in Power BI

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1. Introduction

In this case, we are developing a Power BI dashboard for a logistics company dealing with procurement, ordering, and shipping activities. The objective is to create well-structured visuals to meet the stated reporting requirements, including:

- Slicers for Year, Region, State, Segment, Category, and Subcategory
- A matrix table showing **percentage of regional-wise shipping for different shipping modes** (without filters)
- A matrix table showing **category-wise sales for different shipping modes**
- A trend line visual showing **quantity shipped across months for each shipping mode**
- Later, controlling edit interactions so slicers affect only KPIs and not these tables/charts

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The notes below explain how each requirement is implemented in Power BI, with detailed reasoning for every step and expected outcomes.

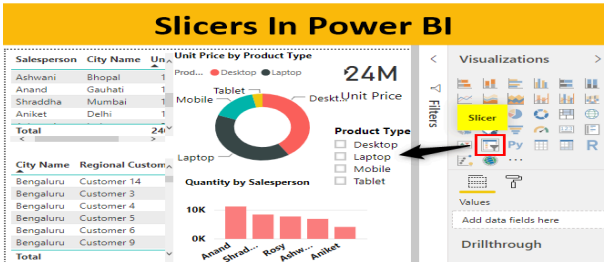
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2. Creating Slicers for Filtering (Requirement Point 3)

Slicers are a crucial feature in Power BI, allowing users to filter visuals interactively. The requirement states:

> "Selection options for Year, Region, State, Segment, Category, and Subcategory must be shown."

We already had the **Year** slicer, so we continue by creating additional slicers.



2.1 Cloning the Year Slicer

To maintain consistent formatting:

1. **Select the Year slicer** on the canvas.
2. Press **Ctrl + C** (copy) and **Ctrl + V** (paste).
3. A duplicate appears in the same position; drag and place it next to the original slicer.

This saves time and ensures:

- Uniform height and width
- Same font sizes
- Same background/border settings
- Same slicer orientation

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2.2 Configuring the Region Slicer

1. Remove the existing field in the cloned slicer (Order Date or Year).
2. Drag **Region** from the fields panel into the slicer's *Field* well.
3. By default, Region may select a single value—this may not reflect all available options.

2.3 Enabling "Select All" and Multiple Selection

To allow users to select multiple regions:

1. Select the slicer → go to **Format Visual → Slicer Settings**.
2. Under **Selection Controls**, turn **Single Select → Off**.
3. This automatically reveals options:
    - **Multi-select with Ctrl**
    - **Select All option**
4. Turn **Select All → On**.

This ensures users can clear all filters or apply all regions at once.

2.4 Adjusting Layout & Alignment

Once the slicer is configured:

- Resize it to visually match the existing slicer
- Align it horizontally with the Year slicer
- Use the arrow buttons in the slicer header to scroll through long lists if needed

2.5 Creating Slicers for Segment, Category, and Subcategory

Repeat the same steps:

1. Copy the Region slicer (since it already has Select All enabled).
2. Change the field in each new slicer:
    - Replace with **Segment**
    - Replace with **Category**
    - Replace with **Subcategory**
3. Resize and align them neatly.

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2.6 Final Layout

With all slicers placed, the report now supports full filtering for:

- **Year** (single select)
- **Region** (multi-select with select all)
- **State**
- **Segment**
- **Category**
- **Subcategory**

2.7 Pending Work — Edit Interactions

The requirement also states:

> "Edit interaction must be controlled only for KPIs and no other charts and tables."

This will be handled **after all visuals are created**, because interaction control must be applied to each visual individually.

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3. Creating Matrix Table for Regional Shipping Percentages (Requirement Point 4)

The fourth requirement specifies:

> "Show the **percentage of region-wise shipping for different shipping modes**, without any filters applied."

A matrix is ideal because:

- Rows = shipping modes
- Columns = regions
- Values = % contribution of each region

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3.1 Adding a Matrix Visual

1. Click on a blank part of the canvas.
2. Click **Matrix** visual from the Visualizations pane.
3. An empty template appears.

3.2 Adding Fields

- **Shipping Mode** → drag into **Rows**
- **Region** → drag into **Columns**
- **Quantity** → drag into **Values**

3.3 Converting Quantity to Percentage of Grand Total

To show percentages instead of raw numbers:

1. In the **Values** section, click the dropdown on **Sum of Quantity**.
2. Choose **Show value as → % of grand total**.

This converts each cell to:

(Total quantity for shipping mode + region) ÷ (Total quantity across dataset)

3.4 Formatting the Matrix for Professional Reporting

a. Increase font size

- Row Headers → 18pt
- Column Headers → 18pt
- Values → 18pt

(Or reduce to 16–17 if space is tight.)

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b. Set table style

Under **Style**, choose a clean layout such as:

- **Condensed**
- **Minimal**

This improves readability and uses space efficiently.

c. Center Alignment

To make all values appear visually consistent:

- Row headers → center
- Column headers → center
- Values → center
- Grand totals → center

d. Apply Borders

Use **Format Painter**:

1. Click a KPI card (which already has borders).
2. Click **Format Painter**.
3. Click the matrix.

Borders enhance structure and give the report a finished look.

3.5 Adding Title

Turn on **Title**, rename it:

> **"Percentage of Region-wise Shipping for Different Modes"**

This makes the visual self-explanatory.

3.6 Validating with Shipping Mode Count

Both quantity and count of shipping mode produce identical percentages because:

- Each row represents shipping mode
- Each row contains all orders for that mode

Hence, using **Count of Shipping Mode** (instead of Sum of Quantity) is more intuitive.

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4. Creating Matrix Table for Category-wise Sales (Requirement Point 5)

The next requirement:

> "Table showing category-wise sales for different shipping modes (no filters applied)."

Because the layout is similar to the previous matrix, the easiest approach is to **copy and modify**.

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4.1 Copying Existing Matrix

1. Select the first matrix.
2. Press **Ctrl + C**, then **Ctrl + V**.
3. Move the new matrix below the first one.

4.2 Changing Fields

- Shipping Mode remains in **Rows**
- Replace **Region → Category** (in Columns)
- Replace **Quantity → Sales** (in Values)

4.3 Removing Decimal Points

Because sales often appear with decimals:

1. Select the Sales values.
2. Go to the top ribbon → set **Decimal Places = 0**.

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This declutters the table.

4.4 Adjusting Font Sizes

If the matrix grows too tall:

- Reduce row/column headers from 18pt → 16–17pt
- Reduce values to 16–17pt

4.5 Updating Title

Rename the title as:

"Category-wise Sales for Different Shipping Modes"

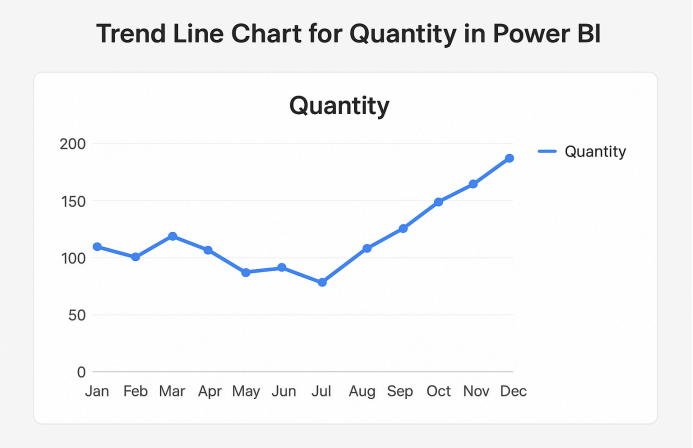
This fulfills Requirement Point 5.

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5. Creating Trend Line Chart for Quantity (Requirement Point 6)

Requirement 6 states:

> "Trend line for quantity for different shipping modes, without any filters applied."



This visual helps identify:

- Monthly shipping patterns
- Peak and low shipping periods
- Seasonal trends
- Comparison of shipping modes

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5.1 Adding a Line Chart

1. Click on the blank canvas.
2. Select **Line Chart** from the Visualizations pane.
3. A blank template appears.

5.2 Adding Fields

a. X-axis — Month

Because trends work best with time-based data:

1. Expand **Order Date** hierarchy.
2. Select **Month**.
3. This produces 12 data points representing Jan to Dec.

b. Y-axis — Quantity

Drag **Quantity** into **Values**.
The line now shows total quantity per month.

c. Legend — Shipping Mode

Drag **Ship Mode** into **Legend**.
This draws separate line paths for:

- First Class
- Same Day
- Second Class
- Standard Class

Each line represents how each shipping mode performed across months.

5.3 Formatting the Trend Line Chart

a. Updating the Title

Rename the title:

"Monthly Quantity by Shipping Mode"

b. Y-axis Label

Under **Y-axis → Title**, rename:

"Quantity Shipped"

This communicates the measurement clearly.

c. Increase Font Size

For readability:

- X-axis → 14–16pt
- Y-axis → 14–16pt
- Legend → 14–16pt
- Title → 16–18pt

d. Applying Border

Use **Format Painter** again to apply the card's border style.

This visually unifies all report visuals.

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6. Understanding “Without Any Filters Applied”

Several requirements mention:

- “without any filters applied”

This means:

- Category table
- Region shipping percentage matrix
- Trend line chart

should **not react** to slicers.

Therefore:

1. Once all visuals (tables, charts, KPIs) are created,
2. We must use **Format → Edit Interactions** to ensure:
    - Slicers affect only KPIs
    - Tables and charts remain static

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This step is intentionally done **at the end**, because:

- You must set interactions **visual by visual**.
- The dashboard must be complete first.

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7. Summary of Concepts and Best Practices Used

7.1 Slicers

- Use single select for Year (to enforce 2021)
- Use multiple select + select all for Region/Segment/etc.
- Slicers help users filter KPIs but not main tables (will configure later)

7.2 Matrix Visuals

Used for comparative analysis across two categorical dimensions:

- Rows (Shipping Modes)
- Columns (Regions or Categories)
- Values (Quantities, Sales, or Percentages)

Formatting makes tables professional and readable.

7.3 Trend Line Chart

Line charts best represent:

- Time-series data
- Monthly/seasonal variations
- Comparative performance across categories (shipping modes)

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7.4 Formatting Techniques

- Font size consistency is key to professional dashboards
- Borders unify the theme
- Center alignment improves clarity
- Titles are important for interpretation
- Decimal places—remove when unnecessary

7.5 Interaction Controls

This is crucial because:

- Some visuals must remain static
- KPIs should react to slicers
- Maintaining correct interactions preserves accuracy

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8. Final Output

By completing these steps, we now have:

1. **Slicers**
2. **Regional-wise shipping percentage matrix**
3. **Category-wise sales matrix**
4. **Monthly trend line for quantity across shipping modes**
5. **Consistent formatting across visuals**

All requirements from points 1 to 6 are fulfilled, except the final interaction controls (to be done after remaining visuals are added).

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